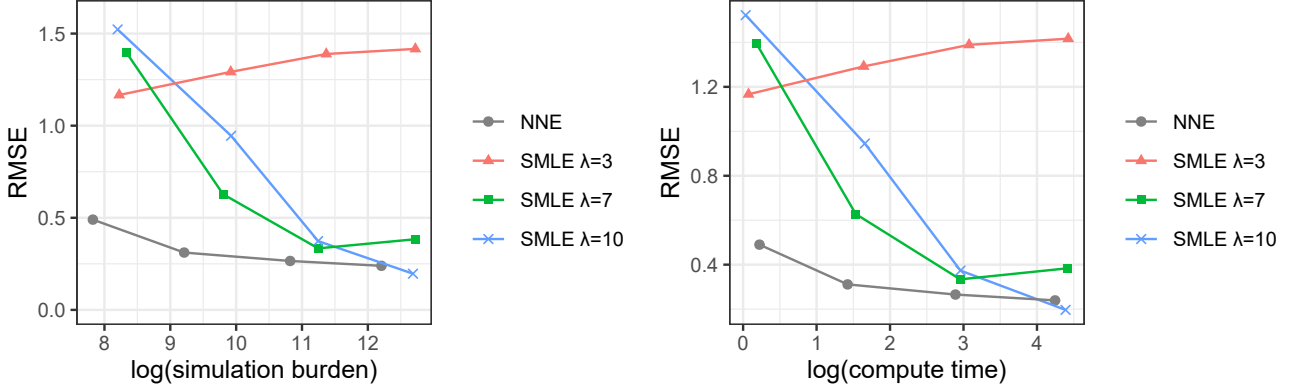




Notes: The “error” in RMSE is defined as the Euclidean distance between the true θ and its estimate. Simulation burden is the number of search model simulations required to carry out estimation. Simulation burden and compute time are varied by changing L^* and R : $L^* \in \{2500, 1e4, 5e4, 2e5\}$, $R \in \{5, 25, 100, 400\}$. The number of hidden nodes increases with L^* and we set it to be roughly $\propto \sqrt{L}$ (see Section 2.4 and Appendix B).

Figure 6: RMSE vs. Computational Cost for Search Model in Monte Carlo

burden, which is the number of search model simulations required to carry out the estimation (for NNE this burden equals L^* ; for SMLE this burden equals R times the number of likelihood evaluations in optimization). The simulation burden is invariant to hardware or code implementation. The right plot of Figure 6 shows the total compute time, with our hardware and code.¹⁵

The results from the two plots in Figure 6 are consistent. NNE has better estimation accuracy than SMLE for a wide range of computation costs, and the advantage is especially large at low computation costs. We see that the estimation accuracy of SMLE is sensitive to λ and the optimal λ changes with computational cost (or R). The sensitivity to λ makes it important to choose the right λ . However, as discussed before, choosing λ is difficult (a comparison like Figure 6 requires repeating SMLE many times as well as knowing the true θ).

Implications by estimates

We examine how the parameter estimates by NNE and SMLE translate into economic predictions about consumer behaviors. We first look at model fit and then counterfactual analysis.

Figure 7 shows the model fit on three key statistics. In each plot, the red dotted line shows the average value observed across 100 Monte Carlo datasets. The histogram shows the values predicted by the search model estimated with these Monte Carlo datasets. The histogram’s spread captures the statistical uncertainty in both $\hat{\theta}$ and the key statistic.

From Figure 7 we see that the predictions under NNE all center around the observed averages. The SMLE again shows sensitivity to the smoothing factor λ , and it shows clear biases under all three smoothing factors. Biases are present even with the optimal $\lambda = 7$ obtained with expensive computations and knowledge of true θ .

¹⁵ A further computational benefit of NNE, not illustrated in Figure 6, is that it is highly parallelizable. All training datasets can be computed at once in parallel, which is useful with computer clusters.